

GENEOmap: trustworthy binary classification of protein binding sites.

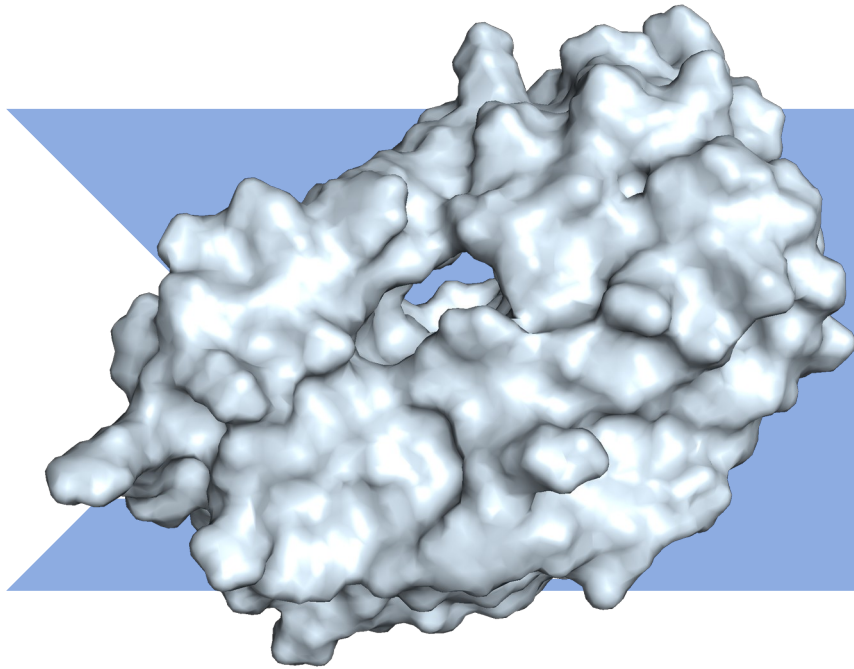
Giovanni Bocchi ¹, Alessandra Micheletti ¹,
Carmen Gratteri ² and Carmine Talarico ²



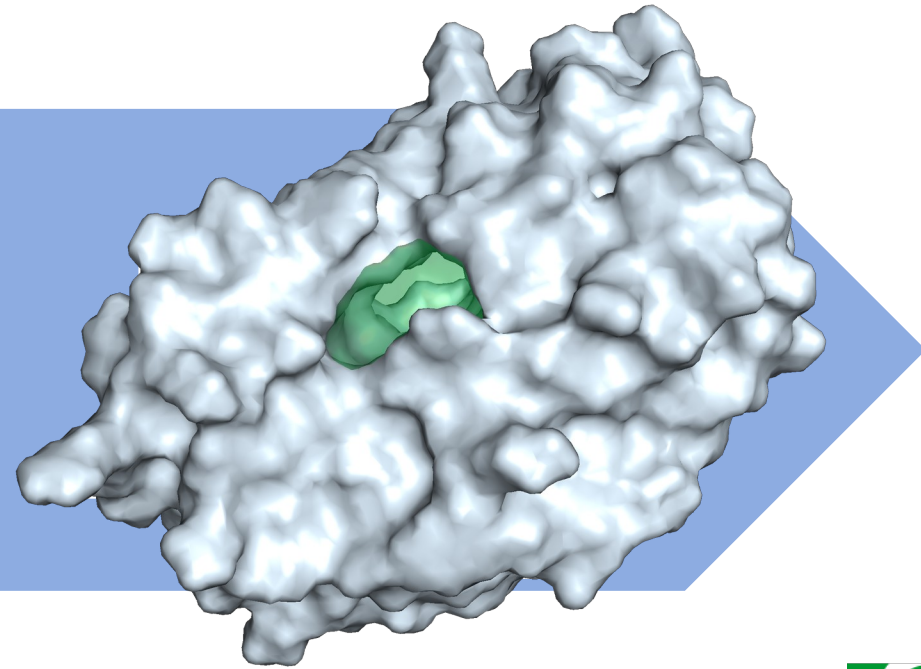
¹ Department of Environmental Science and Policy,
University of Milan

² Dompé Farmaceutici S.p.A.

Problem 1: Protein pocket detection

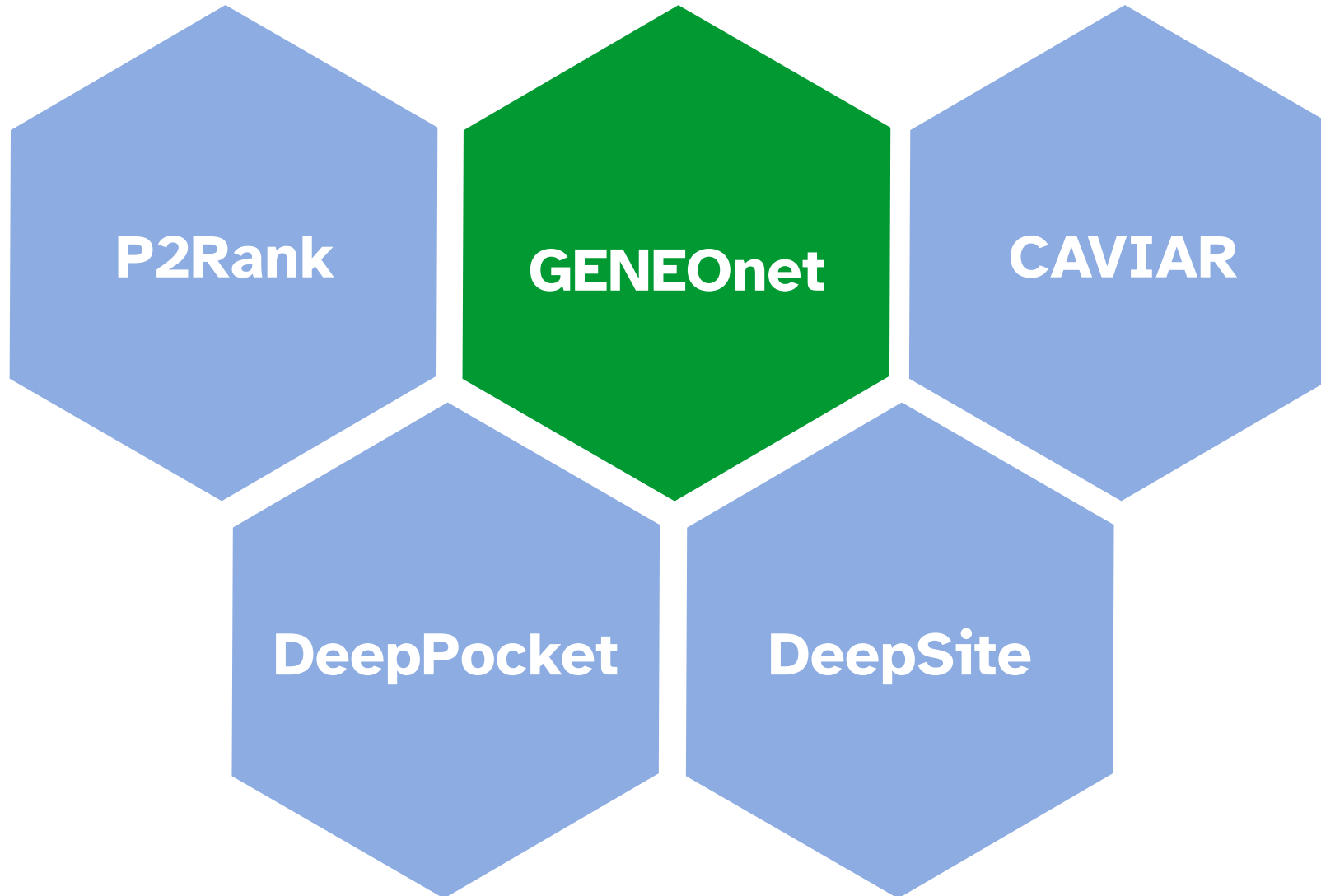


Protein P surface

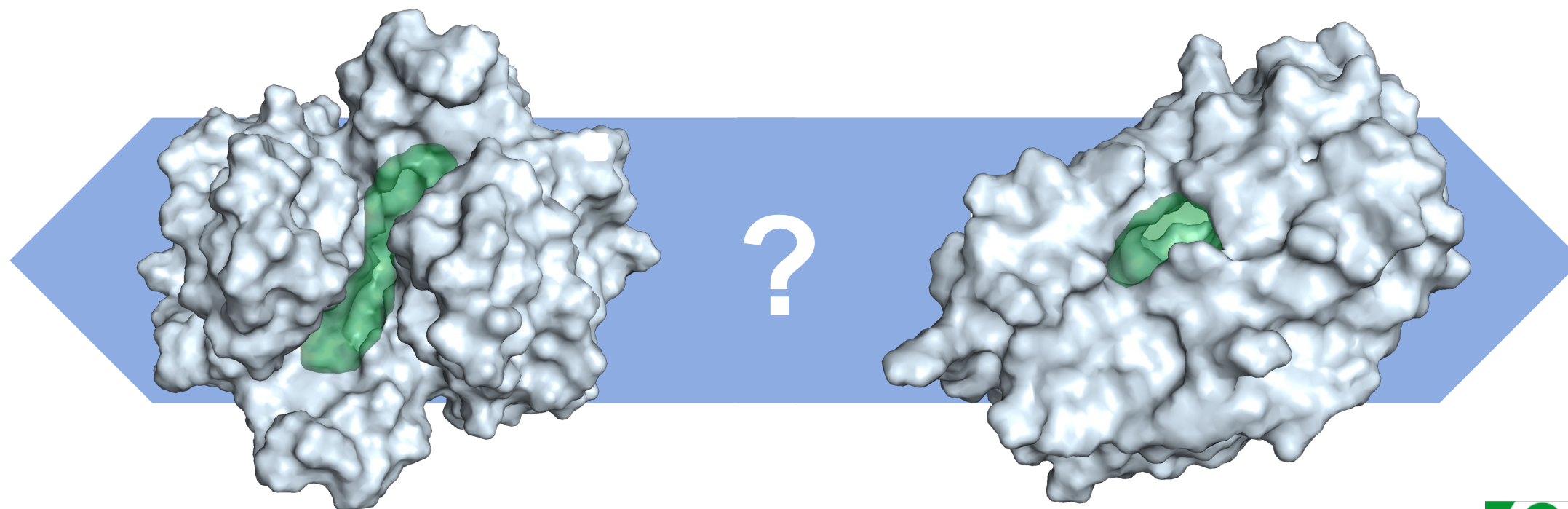


Protein P pocket

Machine learning solutions



Problem 2: Pocket similarity



Protein P pocket

Protein P' pocket

Group Equivariant Non-Expansive Operators

GENEO (Group Equivariant Non-Expansive Operators)

Given two functional spaces $\Phi = \{\varphi: X \rightarrow \mathbb{R}\}$ and $\Psi = \{\psi: Y \rightarrow \mathbb{R}\}$, two groups G and H of transformations of the functions domains (X and Y) and a fixed homomorphism $T: G \rightarrow H$, we define a Group Equivariant Non-Expansive Operator as a function F from Φ to Ψ with the following two properties:

Equivariance: For every $\varphi \in \Phi$ and $g \in G$ it holds that

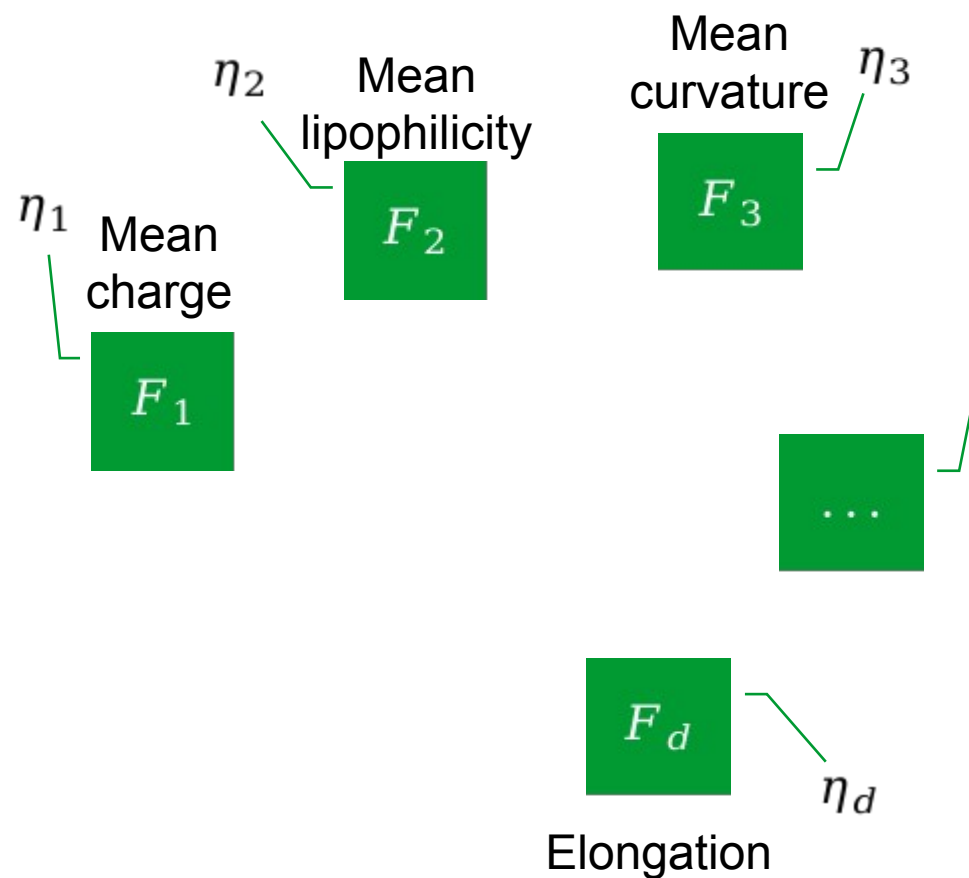
$$F(\varphi \circ g) = F(\varphi) \circ T(g)$$

Non-Expansivity: For every $\varphi_1, \varphi_2 \in \Phi$ it holds that

$$\|F(\varphi_1) - F(\varphi_2)\|_\infty \leq \|\varphi_1 - \varphi_2\|_\infty$$



GENEOmap



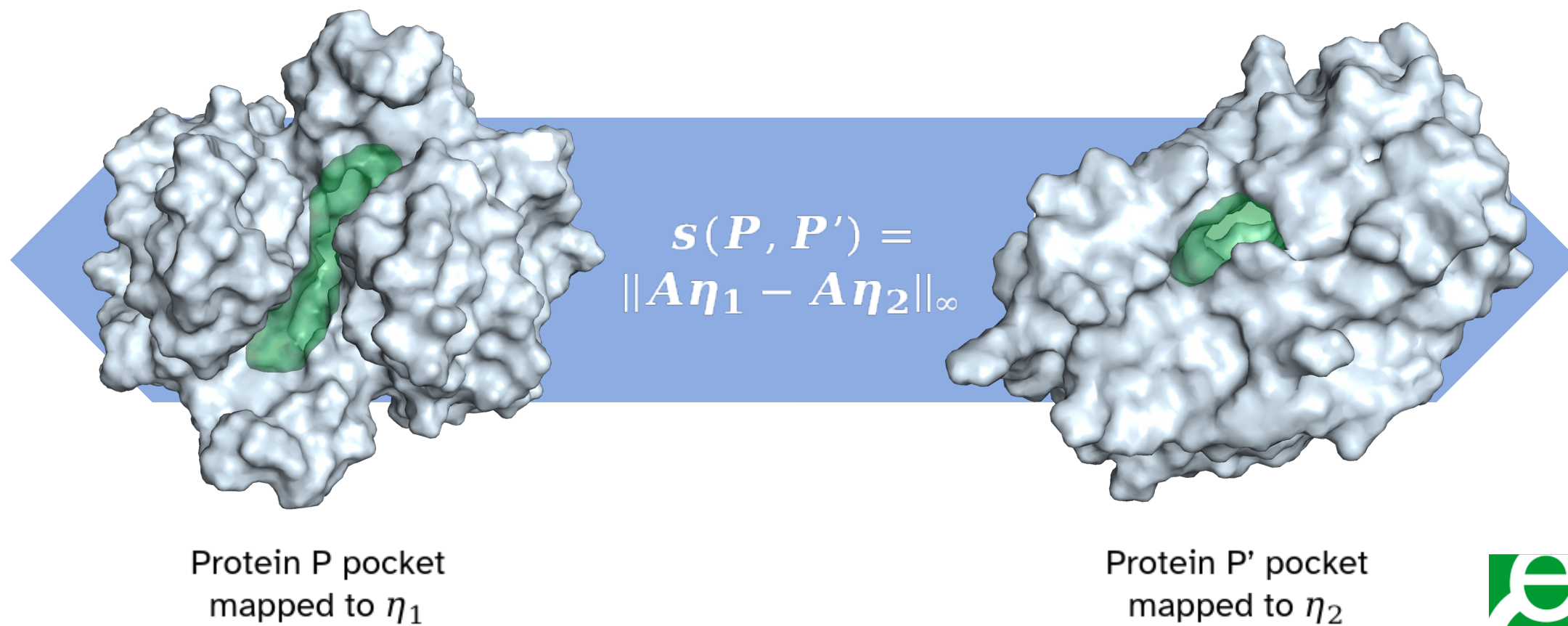
Each GENEO F_i computes an isometry invariant coefficient η_i . Together, such coefficients, constitute the isometry invariant fingerprint of the pocket.

$$\eta = \begin{bmatrix} \eta_1 \\ \vdots \\ \eta_d \end{bmatrix} \in \mathbb{R}^d$$

Lastly, a learnable linear mapping A is applied: $A\eta$, with the requirement $\|A\|_\infty = 1$.



GENEMap: similarity score



GENEMap: binary classification

ProSPECTTs it's a well known benchmark for pocket similarity in a binary classification sense.

Dataset	similar (y=0)	dissimilar (y=1)
D1	13430	92846
D1.2	241	1784
D2	7729	100512
D3	13430	67150
D4	13430	67150
D5	920	5480
D6	19	43
D7	115	56284

Given two pocket embeddings η_1, η_2 we say that they are similar if $\|A\eta_1 - A\eta_2\|_\infty \leq \tau$ or dissimilar otherwise. The matrix A can be learned using a subset of the data.

$$\hat{y} = \mathbf{1}_{[\tau, +\infty)}(\|A\eta_1 - A\eta_2\|_\infty)$$



GENEMap: training

1. For each dataset (D4, D5, D6, D7 from ProSPECCTs and also TOUGH-M1, Vertex), we split the pocket pairs into 80% and 20%.
2. For each dataset, we sampled n similar and n dissimilar pairs from the 80% split.
3. Combining all such data, we obtained the final training set for learning the linear transformation $\mathbf{A}$.
4. We used Adam optimizer for the loss function:

$$\mathcal{L}(\mathbf{A}) = \frac{1}{N} \sum_{i=1}^N (\mathbf{s}(\mathbf{P}, \mathbf{P}')_i - \mathbf{y}_i)^2 - \lambda (\|\mathbf{A}\|_{\infty} - 1)^2$$

where the first term is related to the data, and the second one is a penalty term due to the constraint $\|\mathbf{A}\|_{\infty} = 1$.



Results on ProSPECCTs benchmark

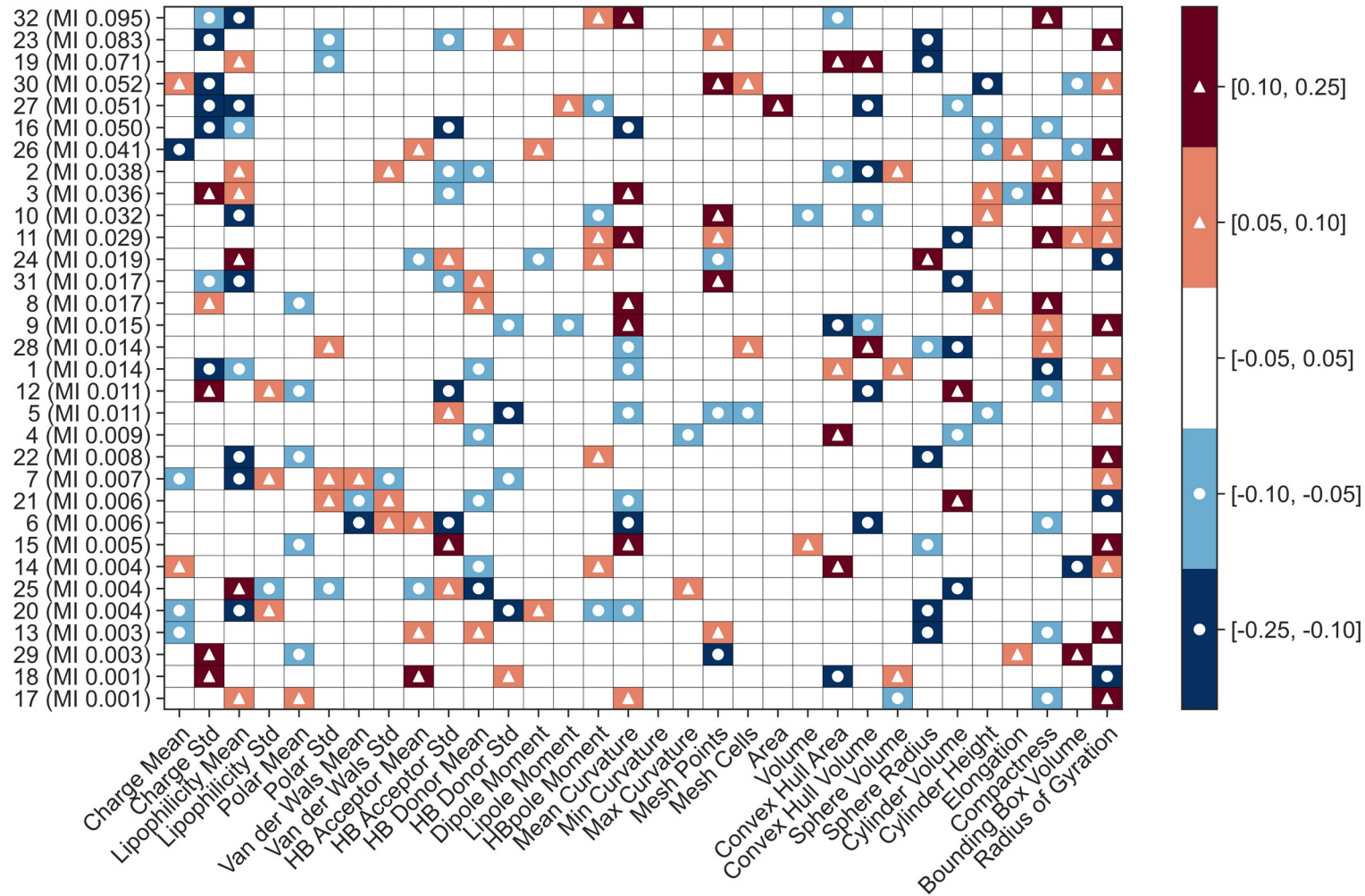
	Model						
Dataset					Site2Vec	DeeplyTough	SiteMine
D1	0.880±0.002		0.879±0.007		1.00	0.95	1.00
D1.2	0.985±0.001		-		0.94	0.98	-
D2	0.930±0.002		0.903±0.002		1.00	0.90	1.00
D3	0.798±0.002		0.836±0.010		0.99	0.76	0.69
D4	0.823±0.002		0.819±0.010		0.99	0.75	0.72
D5	0.929±0.002		0.936±0.005		0.86	0.67	0.78
D6	0.909±0.007		0.885±0.013		0.53	0.54	0.61
D7	0.950±0.002		-		0.66	0.83	0.91
Mean	0.889±0.003		0.876±0.007		0.87	0.80	0.82

GENEOmap has been compared with other methods by computing the AUCs on the ProSPECCTs benchmark.

Considering different scenarios of evaluation, GENEOmap obtains better results than the other methods.



GENEMap: interpretation



After training, we can inspect the structure of matrix A in order to get insights on the classification.

The sparse structure of A helps to further interpret and elucidate the model's functioning.



Thank you for the attention!



Web-page



LinkedIn



Main references

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